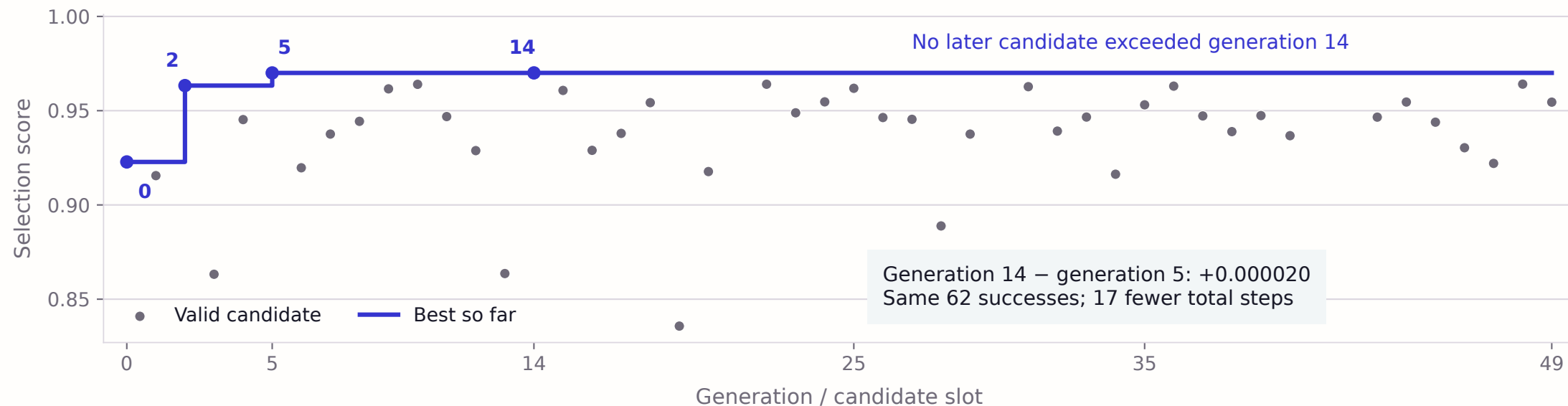


ShinkaDreamer / What evolution actually changed

One campaign · 50 slots · 64 reused development mazes · selected agent: generation 14

A Almost all score improvement arrived by generation 5



B Escape gains remain uncertain

Paired differences in escape rate

Selected – memory baseline

+7.8

Learned – frozen parameters

+3.1

Predictive – fixed-risk planning

+1.6

Escape difference (percentage points)

Frozen: mapping retained, weights fixed.

Fixed-risk: learned forecast use removed from planning.

Intervals: 95% paired episode bootstrap. Development evidence only; selection bias is not corrected.

Better prediction on matched experience is demonstrated here; a general escape benefit is still unestablished.

C Learning improves forecasts

Same actions, worlds, maps and localization

Near-cell Brier loss

3.5% lower than frozen

Threat-cell Brier loss

3.5% lower than frozen

Learned – frozen Brier loss ($\times 10^{-3}$)

Both variants use the same fixed-risk planner.

Negative values indicate better learned forecasts.